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ConstructAI: An Integrated AI-Driven Automated Pre-Construction Feasibility Analysis and Decision Support

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Introduction

The cost of a building is a significant portion of a fixed cost before the first spade is turned. However the land survey, design, approvals, and cost estimate are still being completed individually - as documents, not data. Initiation is the time when the project cost and time are estimated, but it is found that they are consistently exceeded in projects in Sri Lanka (Rathnayake & Ranasinghe, 2020). AI takes care of each task in isolation, with the composition being the missing element.

Objectives

1. Combine site analysis, plan validation, compliance and cost forecasting into one automated pre-construction analysis.
2. Recover parcel boundaries and structural quantities from scanned survey plans and architectural drawings.
3. Screen a design against Urban Development Authority by-laws and return a sequenced approval roadmap.
4. Forecast capital cost and fifty-year maintenance cost from the same geometry.

Method

Four components write into one shared project data model, so a revision in any single analysis propagates to the rest (Fig. 1).

- **C1** — OCR and YOLOv8 read the survey plan; Google Earth Engine grades terrain and flood risk.
- **C2** — YOLOv8 detects walls, doors, windows and columns; scale comes from the dimension annotations (Xu et al., 2024).
- **C3** — by-laws run as parameterised predicates over the shared record (Guo et al., 2021).
- **C4** — a learned BOQ generator prices the works from live market rates; XGBoost (Chen & Guestrin, 2016) predicts service life from thirteen parameters (Taffese & Sistonen, 2017).

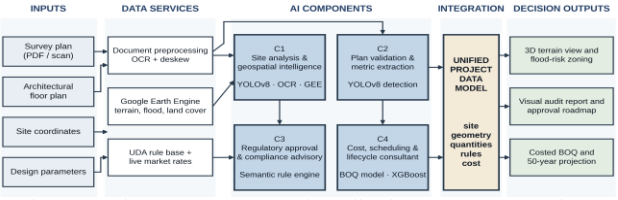


Figure 1 Layered architecture. Every component reads and writes the unified project data model; none calls another directly.

Results and Discussion

Each component is achieved independently on its own scale and is split apart from the other components in a project disjointed manner. The acceptance thresholds are established prior to training and evaluation against them is ongoing (Table 1). The shared record eliminates the manual re-entry between the tools already in place, and a visual audit report and scenario simulator replaces the chatbot.

Table 1 Evaluation metrics and acceptance targets.

Component	What we measure	Target
C1 · Site analysis	Land boundary and buildable area found correctly	mAP 0.90 · area error ≤ 5 %
C2 · Plan validation	Walls, doors and columns found; quantities correct	mAP 0.90 · quantity error ≤ 4 %
C3 · Approval advisory	By-law violations caught, as an expert would	precision 0.95 · recall 0.90
C4 · Cost and lifecycle	Cost close to a surveyor's bill; lifespan realistic	cost error ≤ 10 % · lifespan ± 3 years

Conclusion

ConstructAI's pre-construction feasibility is not four but one analysis. So the geometry that gives the client the capital cost, also gives the client the maintenance cost - which is also based on a clause and a model version - but this time over a 50 year period.

References

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